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Name: _____

May 9, 2015 Saturday

Department/School: _____

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University Physics
Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1.(20pts.) A bullet of mass m is fired straight up out of a gun at ground level and the initial speed of the bullet is v_0 . Assume the air drag on the bullet varies quadratically with its velocity. That is, the resistive force on it is $F_r = -mkv^2$ when the bullet moves up and $F_r = mkv^2$ when the bullet moves down, where k is a constant, $v = \frac{dz}{dt}$ and the positive direction of z is taken to be upwards.

(a) Determine the velocity of the bullet as the function of height during its upward motion. Find the highest point the bullet in terms of v_0 , k and g .

(b) Determine the velocity of the bullet as the function of the distance from its highest point during its downward motion.

(c) Using the results of (a) and (b) to determine the speed of the bullet when it hits the ground in terms of v_0 , k and g .

Solution: (a) The equation of motion for the upward motion is

$$m \frac{dv}{dt} = -mg - mkv^2.$$

Since $\frac{dv}{dt} = \frac{dv}{dz} \frac{dz}{dt} = v \frac{dv}{dz}$, the equation can be rewritten as $v \frac{dv}{dz} = -g - kv^2$ or

$$\frac{d(kv^2)}{kv^2 + g} = -2kdz.$$

Integrating both sides of the equation, we obtain

$$\ln(kv^2 + g) = -2kz + C$$

Since $v = v_0$ when $z = 0$, we have $C = \ln(kv_0^2 + g)$. Therefore

$$v^2 = \left(v_0^2 + \frac{g}{k} \right) e^{-2kz} - \frac{g}{k}.$$

Substituting $v = 0$ into above equation, we have

$$\left(v_0^2 + \frac{g}{k} \right) e^{-2kz_{\max}} = \frac{g}{k}, \quad e^{-2kz_{\max}} = \frac{g}{kv_0^2 + g}, \quad z_{\max} = \frac{1}{2k} \ln \left(\frac{kv_0^2 + g}{g} \right).$$

(b) The equation of motion for the downward motion is

$$m \frac{dv}{dt} = -mg + mkv^2$$

and it can be rewritten as

$$\frac{d(-kv^2)}{g - kv^2} = 2kdz$$

Integrating both sides of the equation we have

$$\ln(g - kv^2) = 2kz + C.$$

Since $v = 0$ when $z = 0$, we have $C = \ln g$. Thus

$$v^2 = (1 - e^{2kz}) \frac{g}{k}.$$

(c) From (b), we know that the velocity of the bullet falling from height $z_{\max}$ is

$$v_f^2 = (1 - e^{-2kz_{\max}}) \frac{g}{k}.$$

Noting that $e^{-2kz_{\max}} = \frac{g}{kv_0^2 + g}$, we have $v_f^2 = \frac{v_0^2 g}{kv_0^2 + g}$.

2. (20pts.) A particle of mass m moves in one dimension along the x-axis. The force acted upon it is given by

$$F(x) = \frac{mc^2}{4} \left(\frac{4a^2}{x^3} - \frac{3a^3}{x^4} - \frac{a}{x^2} \right) (x > 0)$$

where a is a constant of dimension of length and c is the speed of light.

(a) Find the potential energy function of the particle. (Choose the zero potential energy at infinity.)

(b) Find the finite equilibrium points of the particle (excluding the infinity), and identify them as stable or unstable.

(c) If the particle is initially placed at the point $x = 3a$, what is the minimum kinetic energy the particle must have so that the particle can escape to the infinity?

Solution: (a)

$$U(x) = -\int_{\infty}^x F(x) dx = \frac{mc^2}{4} \left(\frac{2a^2}{x^2} - \frac{a^3}{x^3} - \frac{a}{x} \right) (x > 0).$$

(b) From

$$F(x) = \frac{mc^2}{4} \left(\frac{4a^2}{x^3} - \frac{3a^3}{x^4} - \frac{a}{x^2} \right) = \frac{mc^2 a}{4x^4} (4ax - 3a^2 - x^2) = -\frac{mc^2 a}{4x^4} (x - a)(x - 3a) = 0$$

we obtain that the finite equilibrium points of the particle are $x = a$ and $x = 3a$.

For $x = a$, $U''(a) = -F'(a) = -\frac{mc^2}{2a^2} < 0$, the equilibrium is unstable.

For $x = 3a$, $U''(3a) = -F'(3a) = \frac{mc^2}{162a^2} > 0$, the equilibrium is stable.

(c) The potential energy of the particle at $x = 3a$ is $U = -\frac{mc^2}{27}$ and the potential energy at the infinity

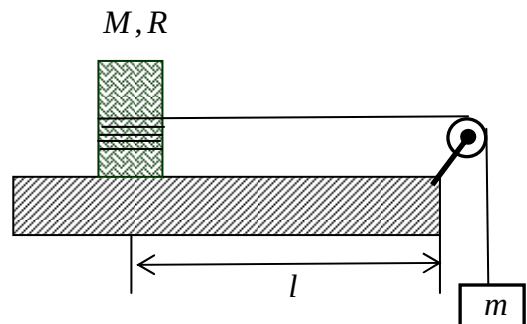
is zero. Thus, to escape to the infinity, the minimum kinetic energy of the particle is $K_{\min} = \frac{mc^2}{27}$.

3. A solid cylindrical spool of mass M and radius R is on the frictionless horizontal table. The spool is connected to a mass m by a massless inextensible string wound around it and through a massless pulley, as shown in the figure. The spool is released from rest and its distance from the edge of the table is l initially. What are the velocity of mass m and the angular velocity of the spool when the center of the spool reaches the edge of the table?

Solution: The equations of rotational and translational motions of the spool can be written as

$$I\ddot{\theta} = TR, \quad (1)$$

$$M\ddot{x} = T \quad (2)$$



where $I = \frac{1}{2}MR^2$. From (1) and (2), we obtain

$$\ddot{x} = \frac{1}{2}R\ddot{\theta}. \quad (3)$$

The equation of motion of mass m is

$$m\ddot{z} = mg - T. \quad (4)$$

Since $\dot{z} = \dot{x} + R\dot{\theta}$, then $\ddot{z} = \ddot{x} + R\ddot{\theta}$. Thus, we have

$$T = mg - m\ddot{x} - mR\ddot{\theta} = mg - 3m\ddot{x} \quad (5)$$

Substituting (5) into (2), we have

$$M\ddot{x} = mg - 3m\ddot{x}.$$

Thus

$$\ddot{x} = \frac{mg}{M+3m}, \quad \ddot{z} = \ddot{x} + R\ddot{\theta} = 3\ddot{x} = \frac{3mg}{M+3m}.$$

From $l = \frac{1}{2}\ddot{x}t^2$, we obtain $t = \sqrt{\frac{2l(M+3m)}{mg}}$. Therefore

$$v = \dot{z}t = \frac{3mg}{M+3m} \sqrt{\frac{2l(M+3m)}{mg}} = \sqrt{\frac{18lmg}{M+3m}},$$

$$\omega = \dot{\theta}t = \frac{2mg}{(M+3m)R} \sqrt{\frac{2l(M+3m)}{mg}} = \sqrt{\frac{8mgl}{(M+3m)R}}.$$

4. (20pts.) The distance from the closest approach to the Sun (perihelion) r_A of a comet of mass m_c is exactly equal to the radius r_E of the Earth's orbit, assumed circular. The speed of the comet at its closest approach to the Sun is $v_A = \sqrt{2}v_E$, where $v_E = \sqrt{GM_S/r_E}$ is the orbital speed of the Earth. (M_S denotes the mass of the Sun.)

(a) What is the total energy of the comet? What type of orbit the comet travels around the Sun?

(b) How long will it take for the comet to move from the closest approach to the Sun (point A) through a angle of 90° , measured from the Sun, to point B?

(Hints: 1. The orbit of a planet or a comet around the Sun can be expressed as $r(\theta) = p/(1 + e \cos \theta)$, where e is the eccentricity. The distance from the closest approach to the Sun is $r_A = r(0) = p/(1 + e)$.

$$2. \int_0^{\pi/2} \frac{d\theta}{(1 + \cos \theta)^2} = \frac{2}{3}.)$$

Solution: (1) The total energy is

$$E = T + U = \frac{1}{2}m_c v_A^2 - \frac{GM_S m_c}{r_A} = m_c v_E^2 - m_c \frac{GM_S}{r_E} = 0.$$

Thus, the orbit is a parabola.

(2) Since the orbit is a parabola, the eccentricity $e=1$, then equation of the trajectory is

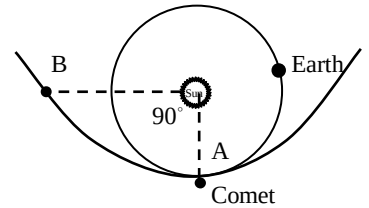
$$r(\theta) = \frac{2r_E}{1 + \cos \theta}.$$

The angular momentum is conserved: $m_c r^2 \dot{\theta} = m_c r_A v_A = m_c \sqrt{2}r_E v_E$.

Then, we obtain

$$\frac{d\theta}{dt} = \frac{\sqrt{2}v_E}{4r_E} (1 + \cos \theta)^2.$$

Thus



$$dt = \frac{2\sqrt{2}r_E}{v_E} \frac{d\theta}{(1 + \cos \theta)^2},$$

$$T = \frac{2\sqrt{2}r_E}{v_E} \int_0^{\frac{\pi}{2}} \frac{d\theta}{(1 + \cos \theta)^2} = \frac{4\sqrt{2}r_E}{3v_E} = \frac{2\sqrt{2}}{3\pi} \frac{2\pi r_E}{v_E} = \frac{2\sqrt{2}}{3\pi} y.$$

5. (20 pts.) A billiard ball A (uniform sphere) of mass m and radius r rolls on a horizontal surface without slipping, with a translational speed v_0 . It collides head-on with billiard ball B elastically. The billiard ball B is also of mass m and radius r and is at rest before the collision. Assume the influences of the frictional forces over the short time duration of collision are negligible.

(a) What are the final translational and rotational velocities of each ball when each ball begins to roll without slipping again after the collision?

(b) What fraction of the mechanical energy is transformed into heat during the process?

Solution: Before the collision

$$v_1 = v_0, \quad \omega_1 = \frac{v_0}{r};$$

$$v_2 = 0, \quad \omega_2 = 0.$$

Just after the head-on elastic collision, we have $v'_{10} = 0$, $v'_{20} = v_0$ (transfer). And the angular momenta about the centers of the ball are conserved, then $\omega'_{10} = \omega_1 = \frac{v_0}{r}$, $\omega'_{20} = \omega_2 = 0$.

The equations of translational and rotational motions of the ball A are

$$m \frac{dv'_1}{dt} = f_1, \quad I \frac{d\omega'_1}{dt} = -f_1 r$$

where $f_1 = \mu mg$, $I = \frac{2}{5}mr^2$. Thus

$$v'_1 = \mu gt \quad \text{and} \quad \omega'_1 = \omega'_{10} - \frac{m\mu gr}{I}t = \frac{v_0}{r} - \frac{5\mu g}{2r}t.$$

The ball begins to roll without slipping when $v'_1 = \omega'_1 r$ or $\mu gt = v_0 - \frac{5\mu g}{2}t$ or $t = \frac{2v_0}{7\mu g}$. Therefore

$$v'_{1f} = \frac{2}{7}v_0, \quad \omega'_{1f} = \frac{2v_0}{7r}.$$

The equations of translational and rotational motions of the ball B are

$$m \frac{dv'_2}{dt} = -f_2, \quad I \frac{d\omega'_2}{dt} = f_2 r$$

$f_2 = \mu mg$, $I = \frac{2}{5}mr^2$. Thus

$$v'_2 = v_0 - \mu gt \quad \text{and} \quad \omega'_2 = \frac{m\mu gr}{I}t = \frac{5\mu g}{2r}t.$$

The ball begins to roll without slipping when $v'_2 = \omega'_2 r$ or $v_0 - \mu gt = \frac{5\mu g}{2}t$ or $t = \frac{2v_0}{7\mu g}$. Therefore

$$v'_{2f} = \frac{5}{7}v_0, \quad \omega'_{2f} = \frac{5v_0}{7r}.$$

(b) The initial kinetic energy is

$$K_i = \frac{1}{2}mv_0^2 + \frac{1}{2}I\left(\frac{v_0}{r}\right)^2 = \frac{7}{10}mv_0^2.$$

The final kinetic energy is

$$K_f = \frac{1}{2}m\left(\frac{2v_0}{7}\right)^2 + \frac{1}{2}I\left(\frac{2v_0}{7r}\right)^2 + \frac{1}{2}m\left(\frac{5v_0}{7}\right)^2 + \frac{1}{2}I\left(\frac{5v_0}{7r}\right)^2 = \frac{203}{490}mv_0^2.$$

The fraction of energy transformed into heat is then

$$\frac{K_i - K_f}{K_i} = \frac{20}{49}.$$

6. (20pts.) A uniform stick of mass M and length L is pivoted at one end.

(a) Find the period of oscillation for small displacements.

(b) If the stick is pivoted about point P a distance x from the center of the mass, what is the period of oscillation for small displacement?

Solution: (a) The equation of rotational motion can be written as

$$I \frac{d^2\theta}{dt^2} = -Mg \frac{L}{2} \sin \theta$$

where $I = \frac{1}{3}ML^2$ is the rotational inertia about the end of the stick. For small displacements, using the approximation $\sin \theta \approx \theta$, the equation can be written as

$$\frac{d^2\theta}{dt^2} + \frac{3g}{2L} \theta = 0.$$

Thus $\omega = \sqrt{\frac{3g}{2L}}$ and $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{2L}{3g}}$.

(b) The equation of rotational motion is

$$I_x \frac{d^2\theta}{dt^2} = -Mgx \sin \theta$$

where $I_x = I_c + Mx^2 = \frac{1}{12}ML^2 + Mx^2$. For small displacements, using the approximation $\sin \theta \approx \theta$, the equation can be written as

$$\frac{d^2\theta}{dt^2} + \frac{gx}{\frac{L^2}{12} + x^2} \theta = 0.$$

Thus $\omega = \sqrt{\frac{gx}{\frac{L^2}{12} + x^2}}$ and $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{\frac{L^2}{12} + x^2}{gx}}$.

